

Consumer Decision-Making Analysis in R

Reproducible, Collaborative, and AI-Aware Workflows

Overview

Structure of the workshop

- ☒ Day 1: The scaffold
- ☐ Day 2: R and data analysis
- ☐ Day 3: Producing a report + Supercharge

The three-day roadmap

Day 1: Scaffold

- R
- Rstudio
- Git/GitHub basics
- Folders
- Data protection
- Quarto

Day 2: Analyse

- Base R
- Import
- Clean
- Transform
- Visualise
- Describe
- Model
- Moderate / mediate
- Collaborate

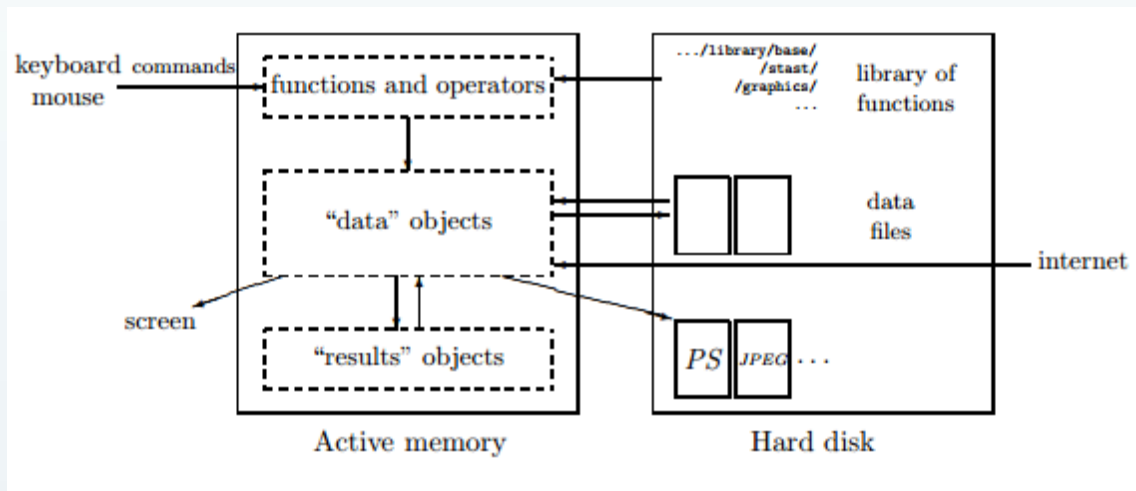
Day 3: Publish

- Export tables and figures
- Write methods
- Write results
- Static/ Annotated/ interactive report
- AI-use statement
- Reproducibility check

Day 2

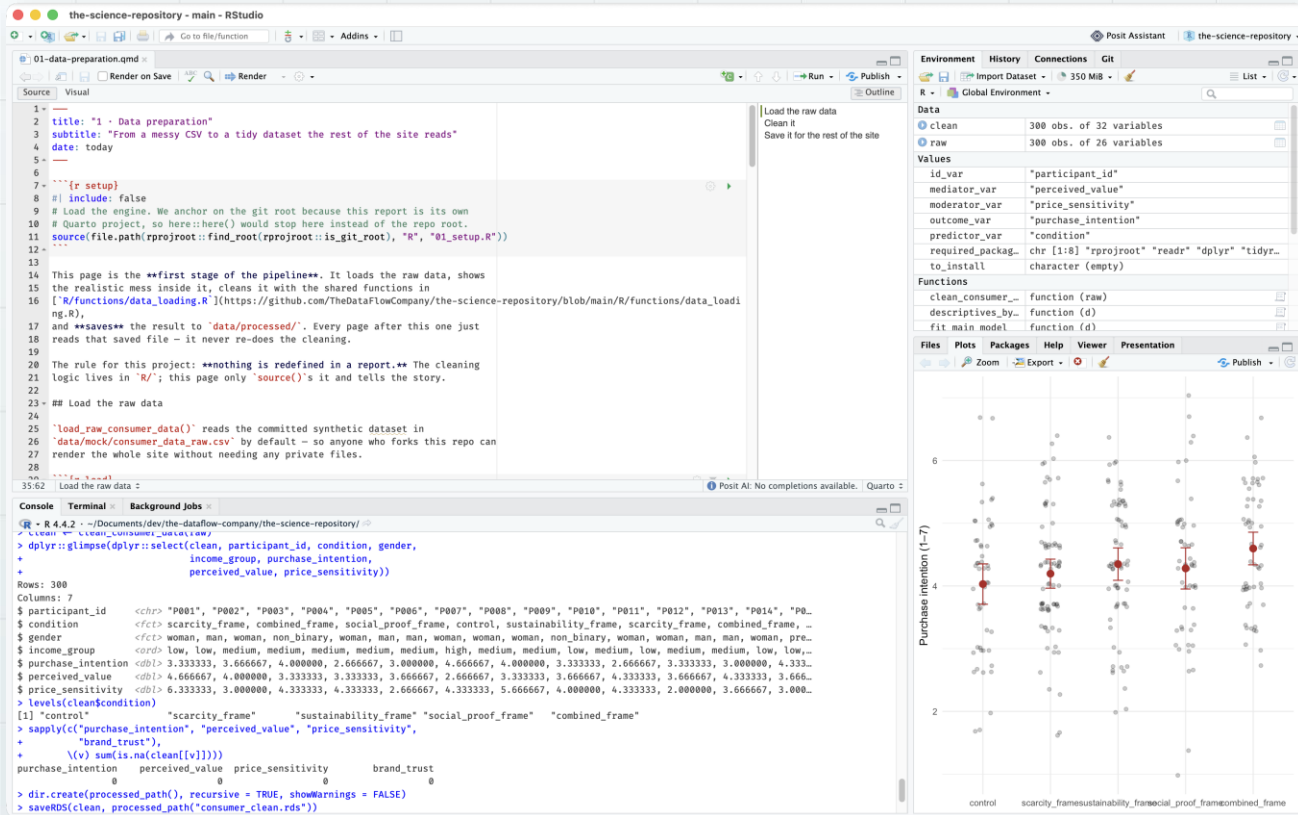
R & Data Analysis

The How



```
where you write  
and save scripts
```

where R runs
R commands



the objects
in memory

```
manage files,  
see figures,  
read docs
```

Git/GitHub

see your changed
and history

R project
setup
your .Rproj
settings

The screenshot shows the RStudio interface with a Quarto report open. The report content includes a title, subtitle, date, and a code chunk for setting up the R environment and loading data. The Environment pane on the right shows the 'Data' section with 'clean' and 'raw' datasets, and the 'Functions' section with various functions like 'clean_consumer_data' and 'descriptives_by'. The Terminal pane at the bottom shows the execution of the code, including the 'dplyr::glance' command and the 'saveRDS' command.

```
1 ---
2 title: "1 • Data preparation"
3 subtitle: "From a messy CSV to a tidy dataset the rest of the site reads"
4 date: today
5 ---
6
7 ```{r setup}
8 #| include: false
9 # Load the engine. We anchor on the git root because this report is its own
10 # Quarto project, so here::here() would stop here instead of the repo root.
11 source(file.path(rprojroot::find_root(rprojroot::is_git_root()), "R", "01_setup.R"))
12
13
14 This page is the first stage of the pipeline. It loads the raw data, shows
15 the realistic mess inside it, cleans it with the shared functions in
16 [ R/functions/data_loading.R ](https://github.com/TheDataFlowCompany/the-science-repository/blob/main/R/functions/data_loading
17 eg.R).
18
19 and saves the result to 'data/processed/'. Every page after this one just
20 reads that saved file – it never re-does the cleaning.
21
22 The rule for this project: nothing is redefined in a report. The cleaning
23 logic lives in 'R/'; this page only 'source()'s it and tells the story.
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26 ## Load the raw data
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```

Terminal
computer's
command line,
'quarto render'
command

R Packages
your packages

Quarto = prose + code → polished output

A .qmd file mixes:

- text (Markdown)
- R code chunks
- the results those chunks produce (tables, figures, numbers)

Quarto runs the code and weaves results into the finished document.

Why Quarto (and not R Markdown)? Quarto is R Markdown's successor:

- one engine for R, Python, and Julia (not R-only)
- one source → HTML, PDF, Word, slides, whole websites
- actively developed, batteries included (citations, cross-references)

Day 3

The Report

Efficient Ins and Outs